Sequence and Series

Section 1: Real Sequences — Foundations & Theory

A thorough understanding of real sequences forms the bedrock of modern real analysis, calculus, and infinite series evaluation.

1.1 Definition and Basic Notation

Definition 1.1: Real Sequence

A **sequence** is defined as a function whose domain is the set of natural numbers $\mathbb{N} = \{1,2,3, \dots\}$ and whose range is a subset of the real numbers $\mathbb{R}$.

Thus, a real sequence a is a set of all ordered pairs $\{(n, a(n))\}_{n=1,2,3,\dots}$.

Since the domain is always the same, a sequence is customarily denoted by $\{a_n\}$ or $\langle a_n \rangle$, where a_n represents the n-th term (general term) of the sequence.

Standard Elementary Examples of Sequences:

Sequence Type	First Terms	General Term Formula (a_n)
Natural Numbers Sequence	1,2,3,4, …	$a_n = n$
Odd Integers Sequence	1,3,5,7,9, …	$a_n = 2n - 1$
Even Integers Sequence	2,4,6,8,10, …	$a_n = 2n$
Alternating Sequence	$1, -1,1, -1,1, -1, \dots$	$a_n = (-1)^{n-1}$ or $(-1)^{n+1}$

1.2 Limit and Convergence Classification

Definition 1.2: Limit of a Sequence

A real sequence $\{a_n\}$ is said to tend to a limit l (written as $\lim_{n\to\infty} a_n = l$) if for every $\epsilon > 0$, there exists a positive integer $N \in \mathbb{Z}^+$ such that:

$$|a_n - l| < \epsilon \quad \text{for all } n > N$$

If l is a finite, unique value, we write $\lim_{n\to\infty} a_n = l$, or $a_n \to l$ as $n \to \infty$.

1. **Convergent Sequence:** The sequence $\{a_n\}$ has a unique, finite limit l.

Examples: $\left\{\frac{1}{n}\right\} \to 0$; $\left\{\frac{2n+1}{1-3n}\right\} \to -\frac{2}{3}$.

2. **Divergent Sequence:** The limit l is infinite ($+\infty$ or $-\infty$). Examples: $\{n\} \to +\infty$; $\{2^n\} \to +\infty$.

3. **Oscillatory Sequence:** The limit is non-unique or alternates indefinitely.

Example1: $\{(-1)^{n-1}\}$ oscillates finitely between -1 and 1;

$\{(-1)^n \cdot n\}$ oscillates infinitely.

Fundamental Limit Algebra Theorems for Sequences:

Theorem Property	Algebraic Expression
Sum Rule	If $\{a_n\} \to l$ and $\{b_n\} \to l^*$, then $\{a_n + b_n\} \to l + l^*$
Constant Multiple Rule	If $\{a_n\} \to l$, then $\{c \cdot a_n\} \to c \cdot l$ (for any constant c)
Product Rule	If $\{a_n\} \to l$ and $\{b_n\} \to l^*$, then $\{a_n \cdot b_n\} \to l \cdot l^*$
Quotient Rule	If $\{a_n\} \to l$ and $\{b_n\} \to l^*$, then $\left\{\frac{a_n}{b_n}\right\} \to \frac{l}{l^*}$ (provided $l^* \neq 0$)

1.3 Boundedness and Monotonicity Theorems

Definition 1.3: Bounded Sequence

A sequence $\{a_n\}$ is said to be **bounded** if there exist real numbers m and M such that $m \leq a_n \leq M$ for every $n \in \mathbb{N}$.

If no such bounds exist, the sequence is **unbounded**.

Example: $\{\sin n\}$ is bounded because $-1 \leq \sin n \leq 1$ for all n.

Definition 1.4: Monotonic Sequence

(i) **Monotonically Increasing:** $a_{n+1} \geq a_n$ for all n (i.e., $a_1 \leq a_2 \leq a_3 \leq \cdots \leq a_n \leq a_{n+1}$).
Example: $\{2^n\}$ is monotonically increasing.

(ii) **Monotonically Decreasing:** $a_{n+1} \leq a_n$ for all n (i.e., $a_1 \geq a_2 \geq a_3 \geq \cdots \geq a_n \geq a_{n+1}$).
Example: $\left\{\frac{1}{n}\right\}$ is monotonically decreasing.

A sequence is **monotonic** if it is either monotonically increasing or monotonically decreasing.

Result 1: Every convergent sequence is bounded.

Important Note: The converse is NOT true. A bounded sequence may not be convergent.

Example: $\{(-1)^n\}$ is bounded between -1 and 1, but it oscillates and does not converge.

Result 2 (Monotone Convergence Theorem): A bounded monotonic sequence is ALWAYS convergent.

Section 2: Sequence Convergence — Problems

2.1 Basic Sequence Convergence Problems

Problem 1: Examine convergence of $\left\{\frac{3n-1}{2n+1}\right\}$

Solution: Given general term $a_n = \frac{3n-1}{2n+1}$. Divide numerator and denominator by n:

$a_n = \frac{3-\frac{1}{n}}{2+\frac{1}{n}}$

Taking the limit as $n \to \infty$: $\lim_{n\to\infty} a_n = \frac{3-0}{2+0} = \frac{3}{2}$.

Conclusion: The sequence is **Convergent** and converges to $\frac{3}{2}$.

Problem 2: Examine convergence of $\left\{\frac{1}{n+(-1)^n}\right\}$

Solution: Given $a_n = \frac{1}{n+(-1)^n}$.

For $n \geq 2$, we have $n - 1 \leq n + (-1)^n \leq n + 1$.

As $n \to \infty$, $n + (-1)^n \to \infty$, which implies $\frac{1}{n+(-1)^n} \to 0$.

Conclusion: The sequence is **Convergent** and converges to 0.

Problem 3: Examine convergence of $\left\{1 + \frac{(-1)^n}{n}\right\}$

Solution: Given $a_n = 1 + \frac{(-1)^n}{n}$. Since $-\frac{1}{n} \leq \frac{(-1)^n}{n} \leq \frac{1}{n}$,

By the Sandwich theorem (Squeeze Theorem,) $\lim_{n\to\infty} \frac{(-1)^n}{n} = 0$.

Taking limit as $n \to \infty$: $\lim_{n\to\infty} a_n = 1 + 0 = 1$.

Conclusion: The sequence is **Convergent** and converges to 1.

Problem 4: Examine convergence of $\left\{\left[\frac{n}{n+1}\right]^2\right\}$

Solution: Given $a_n = \left[\frac{n}{n+1}\right]^2 = \left[\frac{1}{1+1/n}\right]^2$.

Taking limit as $n \to \infty$: $\lim_{n\to\infty} \frac{n}{n+1} = 1$, so $\lim_{n\to\infty} a_n = (1)^2 = 1$.

Conclusion: The sequence is **Convergent** and converges to 1.

Problem 5: Examine convergence of $\{\boldsymbol{sin\ n}\}$

Solution: Given $a_n = \sin n$. As $n \to \infty$, $\sin n$ takes values densely distributed in the interval $[-1,1]$.

Conclusion: The sequence is **Divergent** (Finitely Oscillatory between -1 and 1).

2.2 Problems: Check the convergence of the following sequence with the general term a_n

Problem 1: $a_n = \frac{n^2-2n}{3n^2+n}$

Divide numerator and denominator by n^2: $a_n = \frac{1-2/n}{3+1/n}$. Taking limit as $n \to \infty$: $\lim a_n = \frac{1-0}{3+0} = \frac{1}{3}$. Convergent to $1/3$.

Problem 2: $a_n = 2^n$

As $n \to \infty$, 2^n grows without bound towards $+\infty$. Divergent to $+\infty$.

Problem 3: $a_n = 3 + (-1)^n$

For even n, $a_n = 3 + 1 = 4$. For odd n, $a_n = 3 - 1 = 2$. Divergent (Oscillates between 2 and 4).

Few more problems with solutions:

#	General Term a_n	**Algebraic Transformation / Method**	Limit as $n o \infty$	**Classification**
1	$a_n = \frac{2n+1}{1-3n}$	$\frac{2+1/n}{1/n-3}$	$-\frac{2}{3}$	**Convergent**
2	$a_n = \left[\frac{3n+1}{3n-1}\right]^n$	$\left[\frac{1+1/(3n)}{1-1/(3n)}\right]^n \to \frac{e^{1/3}}{e^{-1/3}}$	$e^{2/3}$	**Convergent**
3	$a_n = \frac{1+(-1)^n}{n}$	Squeeze Theorem: $0 \le a_n \le \frac{2}{n} \to 0$	0	**Convergent**
4	$a_n = (3^{2n+1})^{1/n}$	$3^{\frac{2n+1}{n}} = 3^{2+1/n} \to 3^2$	9	**Convergent**
5	$a_n = \frac{\ln n}{n}$	L'Hopital's Rule: $\lim \frac{1/n}{1} = 0$	0	**Convergent**
6	$a_n = \tanh(n)$	$\frac{1-e^{-2n}}{1+e^{-2n}} \to \frac{1-0}{1+0}$	1	**Convergent**
7	$a_n = 2 + 10(0.1)^n$	$2 + 10(0) = 2$	2	**Convergent**
8	$a_n = \frac{n+(-1)^n}{n}$	$1 + \frac{(-1)^n}{n} \to 1 + 0$	1	**Convergent**

9	$a_n = \frac{2n+1}{1-3\sqrt{n}}$	$\frac{\sqrt{n}(2\sqrt{n}+1/\sqrt{n})}{1/\sqrt{n}-3} \to -\infty$	$-\infty$	**Divergent**
10	$a_n = \frac{n^2-2n+1}{n-1}$	$\frac{(n-1)^2}{n-1} = n-1 \to \infty$	$+\infty$	**Divergent**
11	$a_n = \frac{n}{2^n}$	Exponential grows faster than linear	0	**Convergent**
12	$a_n = \frac{3^n}{n^3}$	Exponential grows faster than polynomial	$+\infty$	**Divergent**
13	$a_n = \frac{(-1)^{n+1}}{2n-1}$	Alternating term divided by growing odd integer	0	**Convergent**
14	$a_n = \frac{1}{(0.9)^n}$	$\left(\frac{1}{0.9}\right)^n = (1.111\ldots)^n \to \infty$	$+\infty$	**Divergent**
15	$a_n = n^{1/n}$	$y = n^{1/n} \Rightarrow \ln y = \frac{\ln n}{n} \to 0 \Rightarrow y \to e^0$	1	**Convergent**

Section 3: Infinite Series & Partial Sums

3.1 Definition and Sequence of Partial Sums

Definition 3.1: Infinite Series and Sequence of Partial Sums

If $u_1, u_2, u_3, \ldots, u_n, \ldots$ is an infinite sequence of real numbers, then the formal expression:

$$u_1 + u_2 + u_3 + \cdots + u_n + \cdots = \sum_{n=1}^{\infty} u_n \quad \text{(or simply } \sum u_n\text{)}$$

is called an **infinite series**.

Let S_n denote the sum of the first n terms: $S_n = u_1 + u_2 + \cdots + u_n = \sum_{k=1}^{n} u_k$. S_n is called the n-th Partial Sum, and $\{S_n\}$ forms the **sequence of partial sums**.

(i) If $S_n \to S$ (a finite unique value) as $n \to \infty$, the series $\sum u_n$ is **CONVERGENT** (Sum $= S$).

(ii) If $S_n \to +\infty$ or $-\infty$ as $n \to \infty$, the series $\sum u_n$ is **DIVERGENT**.

(iii) If S_n does not tend to a unique finite or infinite limit, the series $\sum u_n$ is **OSCILLATORY**.

3.2 The Geometric Series Theorem & Complete Proof

Theorem 3.1: Geometric Series Behavior

Show that the infinite geometric series $1 + r + r^2 + r^3 + \cdots + r^{n-1} + \cdots$:

(i) **Converges** if $|r| < 1$ (with Sum $S = \frac{1}{1-r}$).

(ii) **Diverges** if $r \geq 1$.

(iii) **Oscillates** if $r \leq -1$.

Proof: Let $S_n = 1 + r + r^2 + \cdots + r^{n-1}$. For $r \neq 1$, the finite sum formula is $S_n = \frac{1-r^n}{1-r}$.

Case 1: When $|r| < 1$. As $n \to \infty$, $\lim_{n\to\infty} r^n = 0$. Therefore, $\lim S_n = \lim \frac{1-r^n}{1-r} = \frac{1-0}{1-r} = \frac{1}{1-r}$. Hence, the geometric series **CONVERGES** to $\frac{1}{1-r}$.

Case 2: When $r > 1$. As $n \to \infty$, $\lim_{n\to\infty} r^n = \infty$. $S_n = \frac{r^n-1}{r-1} \to \infty$. Hence, the series **DIVERGES**.

Case 3: When $r = 1$. $S_n = 1 + 1 + 1 + \cdots + 1 = n \to \infty$ as $n \to \infty$. Hence, the series **DIVERGES**.

Case 4: When $r = -1$. The series becomes $1 - 1 + 1 - 1 + 1 - 1 + \cdots$. $S_n = 1$ if n is odd, and $S_n = 0$ if n is even. Since S_n oscillates between 1 and 0, the series is **OSCILLATORY** (Finitely Oscillatory).

Case 5: When $r < -1$. Let $r = -p$ where $p > 1$. Then $r^n = (-p)^n = (-1)^n \cdot p^n$. $S_n = \frac{1-(-1)^n p^n}{1+p} \to -\infty$ if n is even, and $+\infty$ if n is odd. Hence, the series is **OSCILLATORY** (Infinitely Oscillatory).

3.3 Solved Examples on Infinite Series

Example 1: Examine convergence of $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$

This is a geometric series with first term $a = 1$ and common ratio $r = 1/2$. Since $|r| = 1/2 < 1$, the series converges. Sum $S = \frac{a}{1-r} = \frac{1}{1-1/2} = 2$.

Example 2: Examine convergence of $1 - \frac{1}{3} + \frac{1}{3^2} - \frac{1}{3^3} + \cdots$

This is a geometric series with $a = 1$ and $r = -1/3$. Since $|r| = |-1/3| = 1/3 < 1$, the series converges. Sum $S = \frac{a}{1-r} = \frac{1}{1-(-1/3)} = \frac{1}{4/3} = \frac{3}{4}$.

Example 3: Examine convergence of $5 - 4 - 1 + 5 - 4 - 1 + \cdots$

Partial sums: $S_1 = 5, S_2 = 1, S_3 = 0, S_4 = 5, S_5 = 1, S_6 = 0$. The sequence of partial sums $\{S_n\}$ oscillates boundedly among $\{5,1,0\}$. **Conclusion: Oscillatory (Divergent).**

Example 4: Examine convergence of $6 - 10 + 4 + 6 - 10 + 4 + \cdots$

Partial sums: $S_1 = 6, S_2 = -4, S_3 = 0$. $\{S_n\} = \{6, -4, 0, 6, -4, 0, \ldots\}$ oscillates. **Conclusion: Oscillatory (Divergent).**

Example 5 (Telescoping Series): $\frac{1}{1\cdot2}+\frac{1}{2\cdot3}+\frac{1}{3\cdot4}+\cdots$

General term $u_n=\frac{1}{n(n+1)}=\frac{1}{n}-\frac{1}{n+1}$. Partial sum $S_n=\left(1-\frac{1}{2}\right)+\left(\frac{1}{2}-\frac{1}{3}\right)+\cdots+\left(\frac{1}{n}-\frac{1}{n+1}\right)=1-\frac{1}{n+1}$.

Taking limit as $n\to\infty$: $\lim S_n=\lim\left(1-\frac{1}{n+1}\right)=1$. Conclusion: Converges to 1.

Physical Application: Bouncing Ball Total Distance Problem

Problem Statement: Bouncing Ball Physics Model

A ball is dropped from a height h feet above a flat surface. Each time the ball hits the ground after falling a height h, it rebounds a distance $(r\cdot h)$ where $0<r<1$.

Derive the total distance traveled by the ball before coming to rest, and calculate the total distance for:

(i) $h=3$ meters, $r=2/3$

(ii) $h=4$ meters, $r=3/4$

Derivation: The ball first falls distance h. Then it rises (rh) and falls (rh). Then rises (r^2h) and falls (r^2h), and so on.

Total Distance $S=h+2(rh)+2(r^2h)+2(r^3h)+\cdots=h+2hr(1+r+r^2+r^3+\cdots)$.

Using the infinite geometric series sum formula for $|r|<1$: $1+r+r^2+\cdots=\frac{1}{1-r}$.

$$S=h+\frac{2hr}{1-r}=\frac{h(1-r)+2hr}{1-r}=h\cdot\frac{1+r}{1-r}$$

Calculations:

(i) For $h=3, r=2/3$: $S=3\cdot\frac{1+2/3}{1-2/3}=3\cdot\frac{5/3}{1/3}=15$ meters (or feet).

(ii) For $h=4, r=3/4$: $S=4\cdot\frac{1+3/4}{1-3/4}=4\cdot\frac{7/4}{1/4}=28$ meters (or feet).

Section 4: Standard Convergence Tests for Series

4.1 Necessary Condition for Convergence

Theorem 4.1: Necessary Condition for Series Convergence

Let $\sum u_n$ be a series of positive terms. If $\sum u_n$ is convergent, then $\lim_{n\to\infty}u_n=0$.

Proof: Let $S_n=u_1+u_2+\cdots+u_n$. Since $\sum u_n$ is convergent, $\lim_{n\to\infty}S_n=K$ (where K is finite). Also, $\lim_{n\to\infty}S_{n-1}=K$. We know that $u_n=S_n-S_{n-1}$. Taking limits: $\lim_{n\to\infty}u_n=\lim_{n\to\infty}(S_n-S_{n-1})=K-K=0$. **Hence proved.**

Converse Fallacy & Counterexample

Critical Note: The converse is NOT true! If $\lim_{n\to\infty} u_n = 0$, the series $\sum u_n$ MAY OR MAY NOT converge.

Counterexample: Consider the series $1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \cdots + \frac{1}{\sqrt{n}} + \cdots = \sum \frac{1}{\sqrt{n}}$.

Here $u_n = \frac{1}{\sqrt{n}}$, and $\lim_{n\to\infty} u_n = 0$.

However, partial sum $S_n = 1 + \frac{1}{\sqrt{2}} + \cdots + \frac{1}{\sqrt{n}} > n \cdot \frac{1}{\sqrt{n}} = \sqrt{n}$.

As $n \to \infty$, $S_n > \sqrt{n} \to \infty$, so $\sum \frac{1}{\sqrt{n}}$ **DIVERGES** despite $\lim u_n = 0$!

4.2 Preliminary Test for Divergence

Test 4.1: Preliminary Test for Divergence

If $\lim_{n\to\infty} u_n \neq 0$, then the series $\sum u_n$ **MUST DIVERGE**.

Example: Test convergence of $\sum \frac{n}{n+1}$. Here $u_n = \frac{n}{n+1}$, and $\lim_{n\to\infty} u_n = 1 \neq 0$. Since $\lim u_n \neq 0$, the series **DIVERGES**.

4.3 p-Series Test (Harmonic Series of Order p)

Test 4.2: p-Series Test Statement

The infinite harmonic series of order p, denoted $\sum_{n=1}^{\infty} \frac{1}{n^p} = \frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \cdots$:

(i) **CONVERGES** if $p > 1$.

(ii) **DIVERGES** if $p \leq 1$.

Standard Examples of p-Series:

- $\sum \frac{1}{n^2}$: $p = 2 > 1 \Longrightarrow$ **CONVERGES**
- $\sum \frac{1}{\sqrt{n}}$: $p = 1/2 \leq 1 \Longrightarrow$ **DIVERGES**
- $\sum \frac{1}{n}$: $p = 1 \leq 1$ (Harmonic Series) $\Longrightarrow$ **DIVERGES**
- $\sum \frac{1}{n^{3/2}}$: $p = 3/2 > 1 \Longrightarrow$ **CONVERGES**

4.4 Direct Comparison Test

Test 4.3: Direct Comparison Test Statement

Let $\sum u_n$ and $\sum v_n$ be two series of positive real terms.

(i) If $u_n \leq v_n$ for all n, and $\sum v_n$ **CONVERGES**, then $\sum u_n$ also **CONVERGES**.

(ii) If $u_n \geq v_n$ for all n, and $\sum v_n$ **DIVERGES**, then $\sum u_n$ also **DIVERGES**.

Solved Direct Comparison Test Examples:

1. Test $\sum \frac{1}{n!}$: For $n \geq 4$, $n! \geq 2^n \Longrightarrow \frac{1}{n!} \leq \frac{1}{2^n}$. Choose $v_n = \frac{1}{2^n}$ (convergent geometric series). By Comparison Test, $\sum \frac{1}{n!}$ **CONVERGES**.

2. Test $\sum \frac{1}{n^2+1}$: $n^2 + 1 > n^2 \Longrightarrow \frac{1}{n^2+1} < \frac{1}{n^2}$. Choose $v_n = \frac{1}{n^2}$ $(p = 2 > 1)$. By Comparison Test, $\sum \frac{1}{n^2+1}$ **CONVERGES**.

3. Test $\sum \frac{1}{2n-1}$: $2n - 1 < 2n \Longrightarrow \frac{1}{2n-1} > \frac{1}{2n}$. Choose $v_n = \frac{1}{2n} = \frac{1}{2} \cdot \frac{1}{n}$ $(p = 1)$. By Comparison Test, $\sum \frac{1}{2n-1}$ **DIVERGES**.

4. Test $\sum \frac{1}{n \cdot 2^n}$: $n \cdot 2^n \geq 2^n \Longrightarrow \frac{1}{n \cdot 2^n} \leq \frac{1}{2^n}$. Choose $v_n = \frac{1}{2^n}$. By Comparison Test, $\sum \frac{1}{n \cdot 2^n}$ **CONVERGES**.

5. Test $\sum \frac{1}{\sqrt{n}(n+1)}$: $\sqrt{n}(n+1) = n^{3/2} + \sqrt{n} > n^{3/2} \Longrightarrow \frac{1}{\sqrt{n}(n+1)} < \frac{1}{n^{3/2}}$. Choose $v_n = \frac{1}{n^{3/2}}$ $(p = 3/2 > 1)$. By Comparison Test, the series **CONVERGES**.

6. Test $\sum \frac{1}{2^n - 1}$: For $n \geq 1$, $2^n - 1 \geq 2^{n-1} \Longrightarrow \frac{1}{2^n-1} \leq \frac{1}{2^{n-1}}$. Choose $v_n = \frac{1}{2^{n-1}}$ (convergent geometric series). By Comparison Test, $\sum \frac{1}{2^n-1}$ **CONVERGES**.

4.5 Limit Comparison Test

Test 4.4: Limit Comparison Test Statement

Let $\sum u_n$ and $\sum v_n$ be two positive-term series. If:

$$\lim_{n\to\infty} \frac{u_n}{v_n} = l \quad \text{(where } l \text{ is finite and non-zero, } l \neq 0\text{)}$$

then both series $\sum u_n$ and $\sum v_n$ **CONVERGE OR DIVERGE TOGETHER**.

Auxiliary Series Selection Rule: Most often, choose $v_n = \frac{1}{n^p}$ as the auxiliary series, where $p =$ (degree of denominator) $-$ (degree of numerator).

Section 5: Limit Comparison Test

Problem 1: Test convergence of $\sum \frac{1}{n^2+2n}$

Step 1: $u_n = \frac{1}{n^2+2n}$. Highest power of n in denominator is n^2. Choose auxiliary series $v_n = \frac{1}{n^2}$.

Step 2: $\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \frac{n^2}{n^2+2n} = \lim_{n\to\infty} \frac{1}{1+2/n} = 1$.

Step 3: Since $l = 1$ is finite and non-zero ($l \neq 0$), both series behave alike.

Step 4: Since $\sum v_n = \sum \frac{1}{n^2}$ is a p-series with $p = 2 > 1$, $\sum v_n$ CONVERGES.

Conclusion: By Limit Comparison Test, $\sum \frac{1}{n(n+2)}$ **CONVERGES**.

Problem 2: Test convergence of $\sum \frac{n+1}{n^2+2n}$

Step 1: $u_n = \frac{n+1}{n^2+2n}$. Numerator degree $= 1$, denominator degree $= 2$. Choose $v_n = \frac{n}{n^2} = \frac{1}{n}$.

Step 2: $\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \frac{n^2+n}{n^2+2n} = \lim_{n\to\infty} \frac{1+1/n}{1+2/n} = 1$.

Step 3: Since $l = 1 \neq 0$, both series behave alike. Since $\sum v_n = \sum \frac{1}{n}$ diverges ($p = 1$), the given series **DIVERGES**.

Problem 3: Test convergence of $\sum \frac{1}{\ln n}$

Step 1: For $n \geq 2$, $\ln n < n \Longrightarrow \frac{1}{\ln} > \frac{1}{n}$. Choose comparison series $v_n = \frac{1}{n}$.

Step 2: Since $\sum v_n = \sum \frac{1}{n}$ is the divergent harmonic series ($p = 1$), by Direct Comparison Test, $\sum \frac{1}{\ln n}$ **DIVERGES**.

Problem 4: Test convergence of $\sum \frac{\sqrt{2n^2-5n+1}}{4n^3-7n^2+2}$

Step 1: $u_n = \frac{\sqrt{2n^2-5n+1}}{4n^3-7n^2+2}$. Numerator order $\sim n$, denominator order $\sim 4n^3$. Choose $v_n = \frac{n}{n^3} = \frac{1}{n^2}$.

Step 2: $\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \frac{n^3\sqrt{2-5/n+1/n^2}}{n^3(4-7/n+2/n^3)} = \frac{\sqrt{2}}{4}$.

Step 3: Since $l = \frac{\sqrt{2}}{4} \neq 0$, both series behave alike. Since $\sum v_n = \sum \frac{1}{n^2}$ converges ($p = 2 > 1$), the given series **CONVERGES**.

Problem 5: Test convergence of $\sum \frac{1}{n^n}$

Step 1: For $n \geq 2$, $n^n \geq 2^n \Longrightarrow \frac{1}{n^n} \leq \frac{1}{2^n}$. Choose $v_n = \frac{1}{2^n}$ (convergent geometric series).

Conclusion: By Direct Comparison Test, $\sum \frac{1}{n^n}$ **CONVERGES**.

Problem 6: Test convergence of $\sum\tan(1/n)$

Step 1: $u_n = \tan(1/n)$. Choose $v_n = \frac{1}{n}$.

Step 2: $\lim_{n\to\infty} \frac{\tan(1/n)}{1/n} = \lim_{x\to 0} \frac{\tan x}{x} = 1$.

Step 3: Since $l = 1 \neq 0$, both series behave alike. Since $\sum v_n = \sum \frac{1}{n}$ diverges, $\sum\tan(1/n)$ **DIVERGES**.

Problem 7: Test convergence of $\sum \sqrt{n+1} - \sqrt{n-1}$

Step 1: Rationalize numerator: $u_n = \frac{(n+1)-(n-1)}{\sqrt{n+1}+\sqrt{n-1}} = \frac{2}{\sqrt{n+1}+\sqrt{n-1}}$. Choose $v_n = \frac{1}{\sqrt{n}} = \frac{1}{n^{1/2}}$.

Step 2: $\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \frac{2\sqrt{n}}{\sqrt{n+1}+\sqrt{n-1}} = \frac{2}{1+1} = 1$.

Step 3: Since $l = 1 \neq 0$, both series behave alike. Since $p = 1/2 \leq 1$, $\sum v_n$ diverges, so the series **DIVERGES**.

Problem 8: Test convergence of $\sum \frac{1}{\sqrt{n}+\sqrt{n+1}}$

Step 1: Denominator order is $\sqrt{n} = n^{1/2}$. Choose $v_n = \frac{1}{\sqrt{n}}$.

Step 2: $\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \frac{\sqrt{n}}{\sqrt{n}+\sqrt{n+1}} = \frac{1}{1+1} = \frac{1}{2}$.

Step 3: Since $l = 1/2 \neq 0$, both series behave alike. Since $p = 1/2 \leq 1$, the given series **DIVERGES**.

Problem 9: Test convergence of $\frac{1}{1.2.3} + \frac{3}{2.3.4} + \frac{5}{3.4.5} + \cdots \infty$

Step 1: General term $u_n = \frac{2n-1}{n(n+1)(n+2)}$. Numerator degree $= 1$, denominator degree $= 3$. Choose $v_n = \frac{n}{n^3} = \frac{1}{n^2}$.

Step 2: $\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \frac{2n^3-n^2}{n^3+3n^2+2n} = 2$.

Step 3: Since $l = 2 \neq 0$, both series behave alike. Since $\sum v_n = \sum \frac{1}{n^2}$ converges ($p = 2 > 1$), the series **CONVERGES**.

Problem 10: Test convergence of $\frac{1}{4.7.10} + \frac{4}{7.10.13} + \frac{9}{10.13.16} + \cdots \infty$

Step 1: General term $u_n = \frac{n^2}{(3n+1)(3n+4)(3n+7)}$. Numerator degree $= 2$, denominator degree $= 3$. Choose $v_n = \frac{n^2}{n^3} = \frac{1}{n}$.

Step 2: $\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \frac{n^3}{(3n+1)(3n+4)(3n+7)} = \frac{1}{3\cdot 3\cdot 3} = \frac{1}{27}$.

Step 3: Since $l = 1/27 \neq 0$, both series behave alike. Since $\sum v_n = \sum \frac{1}{n}$ diverges ($p = 1$), the series **DIVERGES**.

Problem 11: Test convergence of $1 + \frac{1}{2^2} + \frac{2^2}{3^3} + \frac{3^3}{4^4} + \cdots \infty$

Step 1: General term $u_n = \frac{n^n}{(n+1)^{n+1}} = \frac{1}{n+1} \cdot \left[\frac{n}{n+1}\right]^n = \frac{1}{n+1} \cdot \frac{1}{(1+1/n)^n}$. Choose $v_n = \frac{1}{n}$.

Step 2: $\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \frac{n}{n+1} \cdot \frac{1}{(1+1/n)^n} = 1 \cdot \frac{1}{e} = \frac{1}{e}$.

Step 3: Since $l = 1/e \approx 0.3679 \neq 0$, both series behave alike. Since $\sum v_n$ diverges ($p = 1$), the series **DIVERGES**.

Problem 12: Test convergence of $\frac{\sqrt{2}-1}{3^3-1} + \frac{\sqrt{3}-1}{4^3-1} + \frac{\sqrt{4}-1}{5^3-1} + \cdots$

Step 1: General term $u_n = \frac{\sqrt{n+1}-1}{(n+2)^3-1}$. Numerator order $\sim n^{1/2}$, denominator order $\sim n^3$. Choose $v_n = \frac{n^{1/2}}{n^3} = \frac{1}{n^{5/2}}$.

Step 2: $\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \frac{n^{5/2}\left(\sqrt{n+1}-1\right)}{(n+2)^3-1} = \frac{1}{1} = 1$.

Step 3: Since $l = 1 \neq 0$, both series behave alike. Since $\sum v_n = \sum \frac{1}{n^{5/2}}$ converges ($p = 5/2 > 1$), the series **CONVERGES**.

Statement of Cauchy's n-th Root Test

Cauchy's n-th Root Test (Theorem Statement)

Let $\sum_{n=1}^{\infty} a_n$ be a series with non-negative terms ($a_n \geq 0$ for all n). Suppose the limit

$$L = \lim_{n\to\infty} \sqrt[n]{a_n} = \lim_{n\to\infty} (a_n)^{1/n}$$

exists (or is infinite). Then:

1. **Convergence:** If $L < 1$, the series $\sum_{n=1}^{\infty} a_n$ converges absolutely (and hence converges).
2. **Divergence:** If $L > 1$ or $L = \infty$, the series $\sum_{n=1}^{\infty} a_n$ diverges.
3. **Inconclusive:** If $L = 1$, the test provides no information. The series may converge or diverge.

Generalized Form (Limit Superior): If the ordinary limit does not exist, the test is stated using the limit superior:

$$L = \limsup_{n\to\infty} \sqrt[n]{a_n}$$

The same three conclusions hold.

Proof of the Theorem

Case 1: $L < 1$ (Convergence)

Since $L < 1$, we can choose a real number r such that $L < r < 1$. By the definition of the limit, there exists a positive integer N such that for all $n \geq N$:

$$\sqrt[n]{a_n} < r \implies a_n < r^n$$

The series $\sum_{n=N}^{\infty} r^n$ is a geometric series with common ratio r. Since $0 < r < 1$, this geometric series converges. By the Direct Comparison Test, $\sum_{n=N}^{\infty} a_n$ also converges. Adding the finite sum of the first $N-1$ terms does not affect convergence, so $\sum_{n=1}^{\infty} a_n$ converges.

Case 2: $L > 1$ (Divergence)

Since $L > 1$, there exists a positive integer N such that for all $n \geq N$:

$$\sqrt[n]{a_n} > 1 \implies a_n > 1^n = 1$$

Since $a_n > 1$ for all $n \geq N$, the sequence a_n cannot approach zero as $n \to \infty$ ($\lim_{n\to\infty} a_n \neq 0$). By the n-th Term Test for Divergence, the series $\sum_{n=1}^{\infty} a_n$ diverges.

Essential Calculus Limits

Key Calculus Limits for Root Test

When applying the Root Test, the following standard limits are frequently required:

1. $\lim_{n\to\infty} n^{1/n} = 1 \quad$ and $\quad \lim_{n\to\infty}(P(n))^{1/n} = 1$ for any polynomial $P(n)$.

2. $\lim_{n\to\infty} c^{1/n} = 1$ for any constant $c > 0$.

3. $\lim_{n\to\infty} \left(1 + \frac{x}{n}\right)^n = e^x \implies \lim_{n\to\infty} \left(1 + \frac{x}{n}\right)^{-n} = e^{-x}$.

4. $\lim_{n\to\infty} \left(\frac{n+a}{n+b}\right)^n = e^{a-b}$.

Solved Example Problems

Example 1: Rational Base raised to n-th Power

Test the convergence of the series:

$$\sum_{n=1}^{\infty} \left(\frac{2n+3}{3n+1}\right)^n$$

Solution:

1. Identify the general term: $a_n = \left(\frac{2n+3}{3n+1}\right)^n$.

2. Compute $L = \lim_{n\to\infty}(a_n)^{1/n}$:

$$L = \lim_{n\to\infty} \left[\left(\frac{2n+3}{3n+1}\right)^n\right]^{1/n} = \lim_{n\to\infty} \frac{2n+3}{3n+1} = \frac{2}{3}$$

3. **Conclusion:** Since $L = \frac{2}{3} < 1$, by Cauchy's Root Test, the series **converges**.

Example 2: Exponential Polynomial Ratio

Determine whether the following series converges or diverges:

$$\sum_{n=1}^{\infty} \frac{n^n}{(2n+1)^n}$$

Solution:

1. Express a_n as a single power: $a_n = \left(\frac{n}{2n+1}\right)^n$.

2. Apply the Root Test:

$$L = \lim_{n\to\infty} \sqrt[n]{a_n} = \lim_{n\to\infty} \frac{n}{2n+1} = \frac{1}{2}$$

3. **Conclusion:** Since $L = \frac{1}{2} < 1$, the series **converges**.

Example 3: Series involving e

Test the convergence of the series:

$$\sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)^{-n^2}$$

Solution:

1. Rewrite the general term: $a_n = \left[\left(1 + \frac{1}{n}\right)^n\right]^{-n}$.
2. Take the n-th root and compute the limit:

$$L = \lim_{n\to\infty} \sqrt[n]{a_n} = \lim_{n\to\infty} \frac{1}{\left(1 + \frac{1}{n}\right)^n} = \frac{1}{e} \approx 0.368$$

3. **Conclusion:** Since $L = \frac{1}{e} < 1$, the series **converges**.

Example 4: Logarithmic Base

Test the convergence of the series:

$$\sum_{n=2}^{\infty} \frac{1}{(\ln n)^n}$$

Solution:

1. General term: $a_n = \left(\frac{1}{\ln n}\right)^n$.
2. Apply the Root Test:

$$L = \lim_{n\to\infty} \sqrt[n]{a_n} = \lim_{n\to\infty} \frac{1}{\ln n} = 0$$

3. **Conclusion:** Since $L = 0 < 1$, the series **converges**.

Example 5: Inconclusive Case ($L = 1$)

Demonstrate why the Root Test is inconclusive when $L = 1$ using two different p-series.
Analysis:

- Consider $\sum_{n=1}^{\infty} \frac{1}{n^2}$ (known p-series with $p = 2 > 1$, which **converges**):

$$L = \lim_{n\to\infty} \left(\frac{1}{n^2}\right)^{1/n} = \lim_{n\to\infty} \frac{1}{(n^{1/n})^2} = \frac{1}{1^2} = 1$$

- Consider $\sum_{n=1}^{\infty} \frac{1}{n}$ (harmonic series with $p = 1$, which **diverges**):

$$L = \lim_{n\to\infty} \left(\frac{1}{n}\right)^{1/n} = \lim_{n\to\infty} \frac{1}{n^{1/n}} = \frac{1}{1} = 1$$

Takeaway: Both series yield $L = 1$, yet one converges and the other diverges. Thus $L = 1$ provides no conclusion.

Comparison: Root Test vs. Ratio Test

Feature	Cauchy's Root Test	D'Alembert's Ratio Test
Key Formula	$L = \lim_{n\to\infty}(a_n)^{1/n}$	$L = \lim_{n\to\infty}\left\lvert\frac{a_{n+1}}{a_n}\right\rvert$
Best Used For	Terms with explicit n-th powers, e.g., $(f(n))^n$	Terms involving factorials $n!$ or a^n base powers
Strength	Direct evaluation for powers without simplification	Simplifies ratios of products/factorials
Theoretical Link	If $\lim\left\lvert\frac{a_{n+1}}{a_n}\right\rvert = L$, then $\lim\sqrt[n]{a_n} = L$	Root Test is stronger or equal in applicability

1. Rolle's Theorem

Theorem Statement: Rolle's Theorem

Let $f : [a, b] \to \mathbb{R}$ be a real-valued function defined on a closed interval $[a, b]$ satisfying:

1. $f(x)$ is continuous on the closed interval $[a, b]$.
2. $f(x)$ is differentiable on the open interval (a, b).
3. $f(a) = f(b)$.

Then there exists at least one point $c \in (a, b)$ such that $f'(c) = 0$.

Geometrical Interpretation of Rolle's Theorem

Geometrically, Rolle's theorem states that if a continuous curve $y = f(x)$ starts and ends at the same height ($f(a) = f(b)$) over an interval $[a, b]$, and possesses a well-defined tangent at every point in (a, b), then there exists at least one point $C(c, f(c))$ on the curve between $x = a$ and $x = b$ where the tangent line is horizontal (i.e., parallel to the x-axis, so $f'(c) = 0$).

Practice & Verification Problems for Rolle's Theorem

Verify Rolle's Theorem for each of the following functions on the specified intervals:

Problem 1: $f(x) = \dfrac{\sin x}{e^x}$ on $[0, \pi]$

Solution. 1. $f(x) = e^{-x} \sin x$ is continuous on $[0, \pi]$ and differentiable on $(0, \pi)$.
2. Boundary values: $f(0) = \frac{\sin 0}{e^0} = 0$, and $f(\pi) = \frac{\sin \pi}{e^\pi} = 0 \implies f(0) = f(\pi) = 0$.
3. Derivative: $f'(x) = \frac{\cos x - \sin x}{e^x}$.
4. Setting $f'(c) = 0 \implies \cos c - \sin c = 0 \implies \tan c = 1 \implies c = \pi/4$.
5. Since $c = \pi/4 \in (0, \pi)$, Rolle's Theorem is verified. □

Problem 2: $f(x) = (x - a)^m (x - b)^n$ on $[a, b]$ $(m, n \in \mathbb{Z}^+)$

Solution. 1. Polynomial function, continuous on $[a, b]$ and differentiable on (a, b).
2. $f(a) = 0$ and $f(b) = 0 \implies f(a) = f(b) = 0$.
3. Derivative: $f'(x) = m(x - a)^{m-1}(x - b)^n + n(x - a)^m(x - b)^{n-1} = (x - a)^{m-1}(x - b)^{n-1}[(m + n)x - (mb + na)]$.
4. Setting $f'(c) = 0 \implies (m + n)c - (mb + na) = 0 \implies c = \frac{mb+na}{m+n}$.
5. Since $m, n > 0$, c divides $[a, b]$ internally in ratio $n : m \implies c \in (a, b)$, verifying Rolle's Theorem. □

Problem 3: $f(x) = (x + 2)^3 (x - 3)^4$ on $[-2, 3]$

Solution. 1. Continuous on $[-2, 3]$ and differentiable on $(-2, 3)$.
2. $f(-2) = 0$ and $f(3) = 0 \implies f(-2) = f(3) = 0$.
3. Differentiating: $f'(x) = (x + 2)^2 (x - 3)^3 [3(x - 3) + 4(x + 2)] = (x + 2)^2 (x - 3)^3 (7x - 1)$.
4. Setting $f'(c) = 0 \implies 7c - 1 = 0 \implies c = 1/7$.
5. Since $c = 1/7 \in (-2, 3)$, Rolle's Theorem is verified. □

Problem 4: $f(x) = e^x(\sin x - \cos x)$ on $[\pi/4, 5\pi/4]$

Solution. 1. Continuous on $[\pi/4, 5\pi/4]$ and differentiable on $(\pi/4, 5\pi/4)$.
2. $f(\pi/4) = e^{\pi/4}(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}) = 0$; $f(5\pi/4) = e^{5\pi/4}(-\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}) = 0 \implies f(\pi/4) = f(5\pi/4) = 0$.
3. $f'(x) = e^x(\sin x - \cos x) + e^x(\cos x + \sin x) = 2e^x \sin x$.
4. Setting $f'(c) = 0 \implies 2e^c \sin c = 0 \implies \sin c = 0 \implies c = \pi$.
5. Since $c = \pi \in (\pi/4, 5\pi/4)$, Rolle's Theorem is verified. □

Problem 5: $f(x) = x(x+3)e^{-x/2}$ on $[-3, 0]$

Solution. 1. Continuous on $[-3, 0]$ and differentiable on $(-3, 0)$.
2. $f(-3) = 0$ and $f(0) = 0 \implies f(-3) = f(0) = 0$.
3. $f'(x) = (2x+3)e^{-x/2} - \frac{1}{2}(x^2+3x)e^{-x/2} = -\frac{1}{2}e^{-x/2}(x^2 - x - 6) = -\frac{1}{2}e^{-x/2}(x-3)(x+2)$.
4. Setting $f'(c) = 0 \implies c = -2$ or $c = 3$.
5. Selecting interior point: $c = -2 \in (-3, 0)$. Rolle's Theorem is verified. □

Problem 6: $f(x) = x(x-2)e^{3x/4}$ on $[0, 2]$

Solution. 1. Continuous on $[0, 2]$ and differentiable on $(0, 2)$.
2. $f(0) = 0$ and $f(2) = 0 \implies f(0) = f(2) = 0$.
3. $f'(x) = (2x-2)e^{3x/4} + \frac{3}{4}(x^2 - 2x)e^{3x/4} = \frac{1}{4}e^{3x/4}(3x^2 + 2x - 8) = \frac{1}{4}e^{3x/4}(3x-4)(x+2)$.
4. Setting $f'(c) = 0 \implies 3c - 4 = 0 \implies c = 4/3$.
5. Since $c = 4/3 \in (0, 2)$, Rolle's Theorem is verified. □

Problem 7: $f(x) = \ln\left[\frac{x^2+ab}{x(a+b)}\right]$ on $[a, b]$ $(0 < a < b)$

Solution. 1. Continuous on $[a, b]$ and differentiable on (a, b) as $x > 0$.
2. $f(a) = \ln\left[\frac{a^2+ab}{a(a+b)}\right] = \ln 1 = 0$; $f(b) = \ln\left[\frac{b^2+ab}{b(a+b)}\right] = \ln 1 = 0 \implies f(a) = f(b) = 0$.
3. Rewrite $f(x) = \ln(x^2 + ab) - \ln x - \ln(a+b)$.
4. $f'(x) = \frac{2x}{x^2+ab} - \frac{1}{x} = \frac{x^2-ab}{x(x^2+ab)}$.
5. Setting $f'(c) = 0 \implies c^2 - ab = 0 \implies c = \sqrt{ab}$.
6. Since $a < \sqrt{ab} < b$ (geometric mean lies strictly between a and b), $c \in (a, b)$, verifying Rolle's Theorem. □

Problem 8: $f(x) = 2x^3 + x^2 - 4x - 2$ on $[-1/2, \sqrt{2}]$

Solution. 1. Continuous on $[-1/2, \sqrt{2}]$ and differentiable on $(-1/2, \sqrt{2})$.
2. $f(-1/2) = 2(-1/8) + 1/4 - 4(-1/2) - 2 = 0$; $f(\sqrt{2}) = 2(2\sqrt{2}) + 2 - 4\sqrt{2} - 2 = 0 \implies f(-1/2) = f(\sqrt{2}) = 0$.
3. $f'(x) = 6x^2 + 2x - 4 = 2(3x^2 + x - 2) = 2(3x-2)(x+1)$.
4. Setting $f'(c) = 0 \implies c = 2/3$ or $c = -1$.
5. Selecting interior point: $c = 2/3 \in (-1/2, \sqrt{2})$. Rolle's Theorem is verified. □

Problem 9: $f(x) = \sin^2 x$ on $[0, \pi]$

Solution. 1. Continuous on $[0, \pi]$ and differentiable on $(0, \pi)$.
2. $f(0) = \sin^2 0 = 0$, $f(\pi) = \sin^2 \pi = 0 \implies f(0) = f(\pi) = 0$.
3. Derivative: $f'(x) = 2 \sin x \cos x = \sin(2x)$.
4. Setting $f'(c) = 0 \implies \sin(2c) = 0 \implies 2c = \pi \implies c = \pi/2$.
5. Since $c = \pi/2 \in (0, \pi)$, Rolle's Theorem is verified. □

Non-Applicable Cases (Failure Analysis)

Failure Case 1: Boundary Value Inequality — $f(x) = x^2$ on $[1, 2]$

$f(1) = 1 \neq f(2) = 4$. Condition 3 ($f(a) = f(b)$) fails. Rolle's Theorem is not applicable.

Failure Case 2: Jump Discontinuity — Greatest Integer Function $f(x) = [x]$ on $[-1, 1]$

$f(x)$ has jump discontinuities at integer points ($x = 0, 1$). Condition 1 (continuity) fails.

Failure Case 3: Non-Differentiability — $f(x) = x^{2/3}$ on $[-1, 1]$

$f'(x) = \frac{2}{3x^{1/3}}$ is undefined at $x = 0 \in (-1, 1)$ (vertical tangent). Condition 2 (differentiability) fails.

2. Lagrange's Mean Value Theorem (LMVT)

Theorem Statement: Lagrange's Mean Value Theorem

Let $f : [a, b] \to \mathbb{R}$ be a function such that:

1. $f(x)$ is continuous on the closed interval $[a, b]$.
2. $f(x)$ is differentiable on the open interval (a, b).

Then there exists at least one point $c \in (a, b)$ such that:

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Alternative Interval Form $[a, a + h]$: If f satisfies hypotheses on $[a, a + h]$, then $\exists\, \theta \in (0, 1)$ such that:

$$f(a + h) = f(a) + hf'(a + \theta h)$$

Geometrical Interpretation of LMVT

Geometrically, LMVT states that for any continuous and smooth curve $y = f(x)$ over $[a, b]$, there exists at least one point $C(c, f(c))$ on the curve where the tangent line is parallel to the secant line (chord) connecting boundary points $A(a, f(a))$ and $B(b, f(b))$.

Practice & Verification Problems for LMVT

Problem 1: $f(x) = 2x^2 - 7x + 10$ on $[2, 5]$

Solution. 1. Continuous on $[2, 5]$ and differentiable on $(2, 5)$.
2. $f(2) = 4$, $f(5) = 25 \implies \frac{f(5)-f(2)}{5-2} = \frac{25-4}{3} = 7$.
3. $f'(x) = 4x - 7 \implies 4c - 7 = 7 \implies 4c = 14 \implies c = 7/2 = 3.5$.
4. Since $c = 3.5 \in (2, 5)$, LMVT is verified. □

Problem 2: $f(x) = (x - 1)(x - 2)(x - 3)$ on $[0, 4]$

Solution. 1. $f(x) = x^3 - 6x^2 + 11x - 6$ is continuous on $[0, 4]$ and differentiable on $(0, 4)$.
2. $f(0) = -6$, $f(4) = 6 \implies \frac{f(4)-f(0)}{4-0} = \frac{12}{4} = 3$.
3. $f'(x) = 3x^2 - 12x + 11$.
4. Setting $3c^2 - 12c + 11 = 3 \implies 3c^2 - 12c + 8 = 0 \implies c = 2 \pm \frac{2\sqrt{3}}{3} \approx 2 \pm 1.1547$.
5. Both $c_1 \approx 0.8453 \in (0, 4)$ and $c_2 \approx 3.1547 \in (0, 4)$ verify LMVT. □

Problem 3: $f(x) = \ln x$ on $[1, e]$

Solution. 1. Continuous on $[1, e]$ and differentiable on $(1, e)$.
2. $f(1) = 0$, $f(e) = 1 \implies \frac{f(e)-f(1)}{e-1} = \frac{1}{e-1}$.
3. $f'(x) = 1/x \implies 1/c = \frac{1}{e-1} \implies c = e - 1 \approx 1.7183$.
4. Since $1 < e - 1 < e$, LMVT is verified. □

Problem 4: $f(x) = e^x$ on $[0, 1]$

Solution. 1. Continuous on $[0, 1]$ and differentiable on $(0, 1)$.
2. $f(0) = 1$, $f(1) = e \implies \frac{f(1)-f(0)}{1-0} = e - 1$.
3. $f'(x) = e^x \implies e^c = e - 1 \implies c = \ln(e - 1) \approx 0.5413$.
4. Since $c \approx 0.5413 \in (0, 1)$, LMVT is verified. □

Problem 5: $f(x) = \sin x$ on $[0, \pi]$

Solution. 1. Continuous on $[0, \pi]$ and differentiable on $(0, \pi)$.
2. $f(0) = 0$, $f(\pi) = 0 \implies \frac{f(\pi)-f(0)}{\pi-0} = 0$.
3. $f'(x) = \cos x \implies \cos c = 0 \implies c = \pi/2$.
4. Since $c = \pi/2 \in (0, \pi)$, LMVT is verified. □

Problem 6: $f(x) = x^2$ on $[1, 5]$

Solution. 1. Continuous on $[1, 5]$ and differentiable on $(1, 5)$.
2. $f(1) = 1$, $f(5) = 25 \implies \frac{25-1}{5-1} = 6$.
3. $f'(x) = 2x \implies 2c = 6 \implies c = 3 \in (1, 5)$. LMVT is verified. □

Non-Applicable Cases for LMVT

Failure Case 1: Undefined Point — $f(x) = x^{2/3}$ on $[-1, 1]$

$f'(x) = \frac{2}{3x^{1/3}}$ is undefined at $x = 0 \in (-1, 1)$. Condition 2 (differentiability) fails.

Failure Case 2: Sharp Corner — $f(x) = |x|$ on $[-1, 1]$

$f(x) = |x|$ is non-differentiable at $x = 0 \in (-1, 1)$ ($f'(0^-) = -1 \neq f'(0^+) = +1$). Condition 2 fails.

3. Notes, Deductions & Applications of LMVT

Note 1: Derivation of the Newton-Raphson Method

Suppose $x = a + h$ is an exact root of $f(x) = 0$. Applying LMVT on $[a, a + h]$:

$$0 = f(a + h) = f(a) + hf'(a + \theta h) \quad (0 < \theta < 1)$$

Approximating $f'(a + \theta h) \approx f'(a)$ for small h, we get $h \approx -\frac{f(a)}{f'(a)}$. Iteratively updating x_k gives the Newton-Raphson formula:

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

Note 2: Bounding Theorem for Function Increments

If $m \leq f'(x) \leq M$ for all $x \in (a, b)$, then by LMVT:

$$m \leq \frac{f(b) - f(a)}{b - a} \leq M \implies m(b - a) \leq f(b) - f(a) \leq M(b - a)$$

Note 3: Monotonicity & Constant Function Criteria

Let f satisfy LMVT on $[a, b]$, and let $x_1 < x_2$ be in $[a, b]$. Then $f(x_2) - f(x_1) = (x_2 - x_1)f'(c)$ for some $c \in (x_1, x_2)$:

1. **Constant Function:** If $f'(x) = 0 \ \forall x \in (a, b)$, then $f(x_2) = f(x_1) \implies f(x)$ is constant on $[a, b]$.
2. **Strictly Increasing:** If $f'(x) > 0 \ \forall x \in (a, b)$, then $f(x_2) > f(x_1) \implies f(x)$ is strictly increasing.
3. **Strictly Decreasing:** If $f'(x) < 0 \ \forall x \in (a, b)$, then $f(x_2) < f(x_1) \implies f(x)$ is strictly decreasing.

Problem 1: Arctangent Inequality

Problem: Show that $\frac{\pi}{4} + \frac{3}{25} < \tan^{-1}\left(\frac{4}{3}\right) < \frac{\pi}{4} + \frac{1}{6}$.
Proof: Let $f(x) = \tan^{-1} x$ on $\left[1, \frac{4}{3}\right]$. $f(x)$ is continuous on $\left[1, \frac{4}{3}\right]$ and differentiable on $\left(1, \frac{4}{3}\right)$ with $f'(x) = \frac{1}{1+x^2}$.
By Lagrange's Mean Value Theorem (LMVT), there exists $c \in \left(1, \frac{4}{3}\right)$ such that:

$$f'(c) = \frac{\tan^{-1}(4/3) - \tan^{-1}(1)}{\frac{4}{3} - 1} = \frac{\tan^{-1}(4/3) - \frac{\pi}{4}}{\frac{1}{3}} = 3\left(\tan^{-1}\left(\frac{4}{3}\right) - \frac{\pi}{4}\right)$$

Since $1 < c < \frac{4}{3}$, squaring gives:

$$1 < c^2 < \frac{16}{9} \implies 2 < 1 + c^2 < \frac{25}{9} \implies \frac{9}{25} < \frac{1}{1+c^2} < \frac{1}{2}$$

Substituting $f'(c) = \frac{1}{1+c^2}$:

$$\frac{9}{25} < 3\left(\tan^{-1}\left(\frac{4}{3}\right) - \frac{\pi}{4}\right) < \frac{1}{2}$$

Dividing by 3:

$$\frac{3}{25} < \tan^{-1}\left(\frac{4}{3}\right) - \frac{\pi}{4} < \frac{1}{6}$$

Adding $\frac{\pi}{4}$ throughout:

$$\frac{\pi}{4} + \frac{3}{25} < \tan^{-1}\left(\frac{4}{3}\right) < \frac{\pi}{4} + \frac{1}{6}$$

Problem 2: Arcsine Inequality

Problem: By LMVT, show that $\frac{b-a}{\sqrt{1-a^2}} < \sin^{-1} b - \sin^{-1} a < \frac{b-a}{\sqrt{1-b^2}}$ where $0 < a < b < 1$.
Proof: Let $f(x) = \sin^{-1} x$ on $[a, b]$ where $0 < a < b < 1$. $f(x)$ is continuous on $[a, b]$ and differentiable on (a, b) with $f'(x) = \frac{1}{\sqrt{1-x^2}}$.
By LMVT, there exists $c \in (a, b)$ such that:

$$f'(c) = \frac{\sin^{-1} b - \sin^{-1} a}{b - a} \implies \frac{1}{\sqrt{1-c^2}} = \frac{\sin^{-1} b - \sin^{-1} a}{b - a}$$

Since $a < c < b$, we bound c:

$$a^2 < c^2 < b^2 \implies -a^2 > -c^2 > -b^2 \implies 1 - a^2 > 1 - c^2 > 1 - b^2$$
$$\implies \sqrt{1-a^2} > \sqrt{1-c^2} > \sqrt{1-b^2} \implies \frac{1}{\sqrt{1-a^2}} < \frac{1}{\sqrt{1-c^2}} < \frac{1}{\sqrt{1-b^2}}$$

Substituting $f'(c)$:

$$\frac{1}{\sqrt{1-a^2}} < \frac{\sin^{-1} b - \sin^{-1} a}{b - a} < \frac{1}{\sqrt{1-b^2}}$$

Multiplying through by $(b - a) > 0$:

$$\frac{b-a}{\sqrt{1-a^2}} < \sin^{-1} b - \sin^{-1} a < \frac{b-a}{\sqrt{1-b^2}}$$

Problem 3: Logarithmic Ratio Inequality

Problem: Prove that $\frac{b-a}{b} < \log\left(\frac{b}{a}\right) < \frac{b-a}{a}$ for $0 < a < b$. Hence show that $\frac{1}{4} < \log\left(\frac{4}{3}\right) < \frac{1}{3}$.
Proof: Let $f(x) = \log_e x$ on $[a, b]$ where $0 < a < b$. $f(x)$ is continuous on $[a, b]$ and differentiable on (a, b) with $f'(x) = \frac{1}{x}$.
By LMVT, there exists $c \in (a, b)$ such that:

$$f'(c) = \frac{\log_e b - \log_e a}{b - a} = \frac{\log(b/a)}{b - a} \implies \frac{1}{c} = \frac{\log(b/a)}{b - a}$$

Since $a < c < b$, taking reciprocals gives:

$$\frac{1}{b} < \frac{1}{c} < \frac{1}{a} \implies \frac{1}{b} < \frac{\log(b/a)}{b - a} < \frac{1}{a}$$

Multiplying through by $(b - a) > 0$:

$$\frac{b-a}{b} < \log\left(\frac{b}{a}\right) < \frac{b-a}{a}$$

Deduction: Substitute $b = 4$ and $a = 3$:

$$\frac{4-3}{4} < \log\left(\frac{4}{3}\right) < \frac{4-3}{3} \implies \frac{1}{4} < \log\left(\frac{4}{3}\right) < \frac{1}{3}$$

Problem 4: Arccosine Inequality

Problem: Using LMVT, show that $\frac{\pi}{3} - \frac{1}{5\sqrt{3}} > \cos^{-1}\left(\frac{3}{5}\right) > \frac{\pi}{3} - \frac{1}{8}$.
Proof: Let $f(x) = \cos^{-1} x$ on $[a, b]$ where $0 < a < b < 1$. $f(x)$ is continuous on $[a, b]$ and differentiable on (a, b) with $f'(x) = \frac{-1}{\sqrt{1-x^2}}$.
By LMVT, there exists $c \in (a, b)$ such that:

$$f'(c) = \frac{\cos^{-1} b - \cos^{-1} a}{b - a} \implies \frac{-1}{\sqrt{1 - c^2}} = \frac{\cos^{-1} b - \cos^{-1} a}{b - a}$$

Since $a < c < b$:

$$a^2 < c^2 < b^2 \implies \sqrt{1 - a^2} > \sqrt{1 - c^2} > \sqrt{1 - b^2}$$
$$\implies \frac{-1}{\sqrt{1 - a^2}} > \frac{-1}{\sqrt{1 - c^2}} > \frac{-1}{\sqrt{1 - b^2}}$$
$$\implies \frac{-1}{\sqrt{1 - a^2}} > \frac{\cos^{-1} b - \cos^{-1} a}{b - a} > \frac{-1}{\sqrt{1 - b^2}}$$

Multiplying by $(b - a) > 0$:

$$\frac{a-b}{\sqrt{1 - a^2}} > \cos^{-1} b - \cos^{-1} a > \frac{a-b}{\sqrt{1 - b^2}}$$

Setting $b = 3/5$ and $a = 1/2$ (so $\cos^{-1}(1/2) = \frac{\pi}{3}$):

$$\frac{\frac{1}{2} - \frac{3}{5}}{\sqrt{1 - \frac{1}{4}}} > \cos^{-1}\left(\frac{3}{5}\right) - \frac{\pi}{3} > \frac{\frac{1}{2} - \frac{3}{5}}{\sqrt{1 - \frac{9}{25}}}$$

$$\frac{-1/10}{\sqrt{3}/2} > \cos^{-1}\left(\frac{3}{5}\right) - \frac{\pi}{3} > \frac{-1/10}{4/5} \implies \frac{-1}{5\sqrt{3}} > \cos^{-1}\left(\frac{3}{5}\right) - \frac{\pi}{3} > \frac{-1}{8}$$

Adding $\frac{\pi}{3}$ throughout:

$$\frac{\pi}{3} - \frac{1}{5\sqrt{3}} > \cos^{-1}\left(\frac{3}{5}\right) > \frac{\pi}{3} - \frac{1}{8}$$

Problem 5: Exponential Function Inequality

Problem: Show that for any $x > 0$, $1 + x < e^x < 1 + xe^x$.
Proof: Let $f(t) = e^t$ on $[0, x]$ for $x > 0$. $f(t)$ is continuous on $[0, x]$ and differentiable on $(0, x)$ with $f'(t) = e^t$.
By LMVT, there exists $c \in (0, x)$ such that:

$$f'(c) = \frac{f(x) - f(0)}{x - 0} \implies e^c = \frac{e^x - 1}{x}$$

Since $0 < c < x$, applying monotonic exponential properties:

$$e^0 < e^c < e^x \implies 1 < \frac{e^x - 1}{x} < e^x$$

Multiplying through by $x > 0$:

$$x < e^x - 1 < xe^x$$

Adding 1 throughout:

$$1 + x < e^x < 1 + xe^x$$

Problem 6: Logarithmic Incremental Inequality

Problem: Prove that $\log(1 + x) = \frac{x}{1+\theta x}$ where $0 < \theta < 1$, and hence deduce that $\frac{x}{1+x} < \log(1 + x) < x$ for $x > 0$.
Proof: Let $f(t) = \log(1+t)$ on $[0, x]$ for $x > 0$. Applying LMVT in interval form $f(a+h) = f(a) + hf'(a + \theta h)$ with $a = 0, h = x$:

$$f(x) = f(0) + xf'(\theta x) \quad (0 < \theta < 1)$$

Since $f(t) = \log(1 + t) \implies f'(t) = \frac{1}{1+t}$, we have $f(0) = 0$ and $f'(\theta x) = \frac{1}{1+\theta x}$. Thus:

$$\log(1 + x) = \frac{x}{1 + \theta x}$$

Since $0 < \theta < 1$ and $x > 0$:

$$0 < \theta x < x \implies 1 < 1 + \theta x < 1 + x$$

Taking reciprocals:

$$1 > \frac{1}{1 + \theta x} > \frac{1}{1 + x} \implies \frac{1}{1 + x} < \frac{1}{1 + \theta x} < 1$$

Multiplying through by $x > 0$:

$$\frac{x}{1+x} < \frac{x}{1+\theta x} < x \implies \frac{x}{1+x} < \log(1+x) < x$$

Cauchy's Mean Value Theorem

Let $f(x)$ and $g(x)$ be two functions which are both:

(i) continuous on $[a, b]$

(ii) differentiable on (a, b)

(iii) $g'(x) \neq 0$ for all $x \in (a, b)$

then there exists at least one value $c \in (a, b)$ such that

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}$$

Verification Problems for Cauchy's Mean Value Theorem:

(i) $f(x) = e^x, \quad g(x) = e^{-x} \quad \text{in } [a, b]$

(ii) $f(x) = x^4, \quad g(x) = x^2 \quad \text{in } [a, b]$

(iii) $f(x) = \ln x, \quad g(x) = \frac{1}{x} \quad \text{in } [1, e]$

(iv) $f(x) = \sin x, \quad g(x) = \cos x \quad \text{in } [a, b] \quad \text{where } 0 \leq a < b \leq \frac{\pi}{2}$

(v) $f(x) = \sqrt{x}, \quad g(x) = \frac{1}{\sqrt{x}} \quad \text{in } [a, b], \quad 0 < a < b$

(vi) $f(x) = \frac{1}{x^2}, \quad g(x) = \frac{1}{x} \quad \text{in } [a, b], \quad 0 < a < b$

Taylor's Theorem (Generalized Mean Value Theorem)

Let $f : [a, b] \to \mathbb{R}$ be such that:

(i) $f^{(n-1)}$ is continuous on $[a, b]$

(ii) $f^{(n-1)}$ is differentiable on (a, b) (i.e., $f^{(n)}$ exists in (a, b))

then there exists at least one point $c \in (a, b)$ such that

$$f(b) = f(a) + \frac{b-a}{1!} f'(a) + \frac{(b-a)^2}{2!} f''(a) + \cdots + \frac{(b-a)^{n-1}}{(n-1)!} f^{(n-1)}(a) + \frac{(b-a)^n}{n!} f^{(n)}(c)$$

This is called Taylor's theorem with Lagrange's form of remainder.

Another Form:

If $f : [a, a + h] \to \mathbb{R}$ is such that:

(i) $f^{(n-1)}$ is continuous on $[a, a + h]$

(ii) $f^{(n-1)}$ is differentiable in $(a, a + h)$

then there exists at least one number $\theta \in (0,1)$ such that

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \cdots + \frac{h^n}{n!}f^{(n)}(a+\theta h)$$

Notes:

1. Put $n = 1$, we get Lagrange's Mean Value Theorem.
2. Put $a = 0, h = x$ (taking interval as $[0, x]$):

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \cdots + \frac{x^n}{n!}f^{(n)}(\theta x) \quad (0 < \theta < 1)$$

This is called Maclaurin's theorem with Lagrange form of remainder.

Problem: Verify Taylor's theorem for the function $f(x) = (1-x)^{5/2}$ with Lagrange form of remainder up to 2 terms in the interval $[0, 1]$.

Solution: Let $f(x) = (1-x)^{5/2}$ on $[0, 1]$.

$$f'(x) = -\frac{5}{2}(1-x)^{3/2}, \quad f''(x) = \frac{15}{4}(1-x)^{1/2} \quad \text{exists in } (0,1)$$

$\therefore f, f'$ are continuous and differentiable in $(0, 1)$.

By Taylor's theorem, $\exists\, \theta \in (0,1)$ such that

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2}f''(\theta x)$$

Here $x = 1$, $f(0) = 1$, $f'(0) = -\frac{5}{2}$, $f(1) = 0$, $f''(\theta x) = \frac{15}{4}(1-\theta)^{1/2}$.

$$0 = 1 + 1\left(-\frac{5}{2}\right) + \frac{1^2}{2} \times \frac{15}{4}(1-\theta)^{1/2}$$

$$\frac{3}{2} = \frac{15}{8}(1-\theta)^{1/2} \implies (1-\theta)^{1/2} = \frac{4}{5} \implies 1-\theta = \frac{16}{25} \implies \theta = 1 - \frac{16}{25} = \frac{9}{25} \in (0,1)$$

Hence the theorem is verified.

Taylor's Series

The Taylor's series expansion of $f(x)$ about $x = a$ is:

$$f(x) = f(a) + \frac{x-a}{1!}f'(a) + \frac{(x-a)^2}{2!}f''(a) + \frac{(x-a)^3}{3!}f'''(a) + \ldots$$

The above expansion is also called Taylor's series expansion of $f(x)$ in powers of $(x-a)$.

Note: Put $a = 0$ in the above expansion, we get:

$$f(x) = f(0) + \frac{x}{1!}f'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \ldots$$

This is called Maclaurin's series expansion of $f(x)$ (or the series expansion of $f(x)$ in powers of x).

Problem: Write the series expansion of $\log_e x$ in powers of $(x-1)$ and hence evaluate $\log_e 1.1$ correct to 4 decimal places.

Solution: Let $f(x) = \log_e x$.

$$f(1) = 0, \quad f'(x) = \frac{1}{x} \implies f'(1) = 1, \quad f''(x) = -\frac{1}{x^2} \implies f''(1) = -1$$

$$f'''(x) = \frac{2}{x^3} \implies f'''(1) = 2, \quad f^{IV}(x) = -\frac{6}{x^4} \implies f^{IV}(1) = -6$$

The Taylor's series expansion of $f(x)$ in powers of $(x-a)$ is:

$$f(x) = f(1) + \frac{x-1}{1!}f'(1) + \frac{(x-1)^2}{2!}f''(1) + \frac{(x-1)^3}{3!}f'''(1) + \frac{(x-1)^4}{4!}f^{IV}(1) + \dots$$

$$\log_e x = 0 + (x-1)(1) + \frac{(x-1)^2}{2!}(-1) + \frac{(x-1)^3}{3!}(2) + \frac{(x-1)^4}{4!}(-6) + \dots$$

$$\log_e x = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \frac{(x-1)^4}{4} + \dots$$

Now put $x = 1.1 \implies x - 1 = 0.1$:

$$\log_e 1.1 \approx 0.1 - \frac{(0.1)^2}{2} + \frac{(0.1)^3}{3} - \frac{(0.1)^4}{4} = 0.1 - 0.005 + 0.000333 - 0.000025 = 0.0953$$

Problem: Expand $\sin x$ in powers of $(x - \frac{\pi}{2})$ and hence find the value of $\sin 91^\circ$ correct to 4 decimal places.

Solution: Let $f(x) = \sin x$.

$$f\left(\frac{\pi}{2}\right) = 1, \quad f'(x) = \cos x \implies f'\left(\frac{\pi}{2}\right) = 0, \quad f''(x) = -\sin x \implies f''\left(\frac{\pi}{2}\right) = -1$$

$$f'''(x) = -\cos x \implies f'''\left(\frac{\pi}{2}\right) = 0, \quad f^{IV}(x) = \sin x \implies f^{IV}\left(\frac{\pi}{2}\right) = 1$$

The Taylor's series expansion of $f(x)$ in powers of $(x-a)$ is:

$$\sin x = 1 + \left(x - \frac{\pi}{2}\right)(0) + \frac{(x-\frac{\pi}{2})^2}{2}(-1) + \frac{(x-\frac{\pi}{2})^3}{6}(0) + \frac{(x-\frac{\pi}{2})^4}{24}(1) + \dots$$

$$\sin x = 1 - \frac{1}{2}\left(x - \frac{\pi}{2}\right)^2 + \frac{1}{24}\left(x - \frac{\pi}{2}\right)^4 - \dots$$

Put $x = 91^\circ \implies x - \frac{\pi}{2} = 91 \times \frac{\pi}{180} - 90 \times \frac{\pi}{180} = \frac{\pi}{180} \approx 0.0174$.

$$\sin 91^\circ \approx 1 - \frac{1}{2}(0.0174)^2 + \frac{1}{24}(0.0174)^4 \approx 1 - 0.000151 = 0.9998 \,(\text{or } 0.9999)$$

Problem: Using Taylor's theorem, express the polynomial $2x^3+7x^2+x-6$ in powers of $(x-1)$.

Solution: Let $f(x) = 2x^3 + 7x^2 + x - 6 \implies f(1) = 4$.

$$f'(x) = 6x^2 + 14x + 1 \implies f'(1) = 21$$

$$f''(x) = 12x + 14 \implies f''(1) = 26$$

$$f'''(x) = 12 \implies f'''(1) = 12$$

By Taylor's series expansion of $f(x)$ about $(x-a)$:

$$f(x) = f(1) + \frac{x-1}{1!}f'(1) + \frac{(x-1)^2}{2!}f''(1) + \frac{(x-1)^3}{3!}f'''(1)$$

$$f(x) = 4 + 21(x-1) + 13(x-1)^2 + 2(x-1)^3$$

Problem: Compute to 4 decimal places, the value of $\cos 32^\circ$ by the use of Taylor's series.

Solution: Let $f(x) = \cos x$. Expanding about $x = \frac{\pi}{6}$:

$$f\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}, \quad f'\left(\frac{\pi}{6}\right) = -\frac{1}{2}, \quad f''\left(\frac{\pi}{6}\right) = -\frac{\sqrt{3}}{2}, \quad f'''\left(\frac{\pi}{6}\right) = \frac{1}{2}, \quad f^{IV}\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$$

The Taylor series expansion of $f(x)$ about $x = \frac{\pi}{6}$ is:

$$\cos x = \frac{\sqrt{3}}{2} + \left(x - \frac{\pi}{6}\right)\left(-\frac{1}{2}\right) + \frac{(x-\frac{\pi}{6})^2}{2}\left(-\frac{\sqrt{3}}{2}\right) + \frac{(x-\frac{\pi}{6})^3}{6}\left(\frac{1}{2}\right) + \frac{(x-\frac{\pi}{6})^4}{24}\left(\frac{\sqrt{3}}{2}\right) + \dots$$

Put $x = 32^\circ \implies x - \frac{\pi}{6} = 32 \times \frac{\pi}{180} - 30 \times \frac{\pi}{180} = \frac{2\pi}{180} = \frac{\pi}{90} \approx 0.0349$.

$$\cos 32^\circ = \frac{\sqrt{3}}{2} - \frac{1}{2}(0.0349) - \frac{\sqrt{3}}{4}(0.0349)^2 + \frac{1}{12}(0.0349)^3 \approx 0.8660 - 0.0174 - 0.0005 = 0.8461$$

Problem: Expand e^x by Maclaurin's series.

Solution: Let $f(x) = e^x \implies f(0) = 1, f'(0) = 1, f''(0) = 1, f'''(0) = 1$. By Maclaurin's series expansion:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

Problem: Expand $e^{\sin x}$ by Maclaurin's series expansion up to the terms containing x^4.

Solution: Let $f(x) = e^{\sin x} \implies f(0) = 1$.

$$f'(x) = e^{\sin x}\cos x \implies f'(0) = 1$$

$$f''(x) = e^{\sin x}(\cos^2 x - \sin x) \implies f''(0) = 1$$

$$f'''(x) = e^{\sin x}(-\sin 2x - \cos x + \cos^3 x - \sin x \cos x) \implies f'''(0) = 0$$

$$f^{IV}(0) = -3$$

By Maclaurin's series expansion:

$$e^{\sin x} = 1 + x + \frac{x^2}{2} - \frac{x^4}{8} + \dots$$